

ANDREW CRIX

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Summary

Analytical and driven engineering physics student with strong quantitative, critical thinking, and communication skills. Eager to apply engineering principles to contribute to solutions that address global issues.

Education

B.A.Sc: Engineering Physics | Final-Year GPA: 4.19/4.3

Expected in 06/2026

Queen's University

Kingston, ON

- Relevant Coursework:** High Performance Computational Physics (Grad Class); Mobile Robotics; Automatic Control; Stochastic Control and Optimization (Grad Class-Self Study); Electronics; Mechatronics; Differential Equations; Linear Algebra; Advanced Engineering Physics Design
- Scholarships | Awards:** Dean's Scholar; OUA Academic All Star; Most Improved Athlete; NSERC USRA; Robert Tyre Jones Jr Scholarship; Harold E. Johns Studentship; Loran Scholarship Semi Finalist; Support Our Troops National Scholarship; R. H. Clark Applied Science Centennial Scholarship

Academic Exchange: Physics And Economics

05/2023

University of St. Andrews

St. Andrews, Scotland

Skills

Programming: Python, C++, SQL, MATLAB, PHP, TypeScript/JavaScript, HTML5, CSS, LabVIEW

Libraries/Tools: ROS 2, SciPy, scikit-learn, pandas, PyTorch, TensorFlow, CVXPY, NumPy, mpi4py, Django REST Framework, Git, AWS, Bash, Linux, DAQ

Strengths: Data Structures & Algorithms, Object-Oriented Programming, Machine Learning, Optimization and Control, Robotics, Adaptability, Collaboration

Projects

EEG Foundation Model for Heterogeneous Sensor Layouts

- Introduced graph neural networks to handle varying electrode layouts and a neuroscience-informed loss. Beat previous models accuracy by 10%.
- Won the "Spotlight Paper Award" for most outstanding paper and presentation at Canada's top undergraduate conference on AI, CUCAI.

Hybrid State Estimation using Physics Informed Neural Networks

- Built a hybrid PINN-UKF algorithm for nonlinear robotic systems, improving state estimation accuracy under simulated disturbances.
- Developing a physical differential-drive robot running ROS 2 to integrate the algorithm, with physics-based ensemble training to reduce overfitting.

Work Experience

Combat Medic (25 Field Ambulance Reserve Unit)

04/2025 to Current

Canadian Armed Forces (Royal Canadian Medical Service)

Toronto, Ontario

- Medical support for 32 Canadian Brigade Group. Displays integrity, stewardship and excellence in domestic and international theatres.

Founder/Private Piano Teacher

05/2020 to Current

Self-Employed

Toronto, ON | Remote, ON

- Teach personalized piano lessons. Built a Django-based piano learning platform with 200+ lessons, integrating CI/CD deployment workflows.
- Utilized machine learning techniques for pitch recognition and personalized recommendations to support adaptive online music education.

Research Assistant (NSERC USRA)

05/2024 to 08/2024

Queen's University - Dr Jackson Crane

Kingston, Ontario

- Developed a detonation flow visualization and data acquisition system, now the primary diagnostic at Crane Energy Group.
- Designed and validated a multilayer high-speed PCB with sub-nanosecond edge-rate capability, enabling reliable LED pulsing.
- Synchronized camera, LED pulsing, ignition, and pressure sensing via DAQ, enabling 100,000 fps imaging with orders-of-magnitude less blurring.

Program Assistant

06/2023 to 08/2023

SHAD Canada

St. John's, NL

- Helped deliver a rewarding program for 64 participants. Built a python based synthesizer for my guest lecture on music synthesis and Fourier series.

Software Engineer Intern

05/2022 to 08/2022

Cancer Care Ontario - Sunnybrook Hospital

Toronto, Ontario

- Architected a modular PHP platform for user, document, and communication tracking. Used as the department tracking software after my departure.
- Designed secure backend systems with session-based authentication, permission controls, validation, and secure database querying.

Volunteer/Extracurriculars

Queen's University | Kingston, ON | 2020 - 2026

- Events:** Orientation week leader, QMIND, Queen's Space Conference Co-Chair, "How to Change the World" (Most Implemental Award Winner), Varsity Athlete (Rowing), Effective Altrismus Society Member, The Knowledge Society (1st Cohort)

Global Medic | Toronto, ON | 2021 - current

- Current local volunteer with future goal of international deployment with the Rapid Response Team.

Epilepsy Toronto | Toronto, ON | 2013 - current

- Presentations for community groups and schools; helped with local fundraising events and lead an Accessibility team.